# CSE 400: Fundamentals of Probability in Computing
# Tutorial–2 Lecture Scribe

## Q1. Random Point on a Line Segment

Let a point be chosen at a distance $x$ from one end of a segment of length $L$, where

$$0 < x < L.$$

Two segments are formed: $x$ and $L - x$. Define

$$R = \frac{\min(x, L - x)}{\max(x, L - x)}.$$

We compute $P\left(R < \frac{1}{4}\right)$.

### Case 1: $x \le \frac{L}{2}$

$$\begin{aligned} \frac{x}{L - x} &< \frac{1}{4} \\ 4x &< L - x \\ 5x &< L \\ x &< \frac{L}{5}. \end{aligned}$$

### Case 2: $x > \frac{L}{2}$

$$\begin{aligned} \frac{L - x}{x} &< \frac{1}{4} \\ 4(L - x) &< x \\ 4L - 4x &< x \\ 5x &> 4L \\ x &> \frac{4L}{5}. \end{aligned}$$

Valid region:

$$x \in \left(0, \frac{L}{5}\right) \cup \left(\frac{4L}{5}, L\right).$$

Total valid length:

$$\frac{L}{5} + \frac{L}{5} = \frac{2L}{5}.$$

Since uniform:

$$P = \frac{2L/5}{L} = \frac{2}{5}.$$

## Q2. Bayesian Classification with Normal Distributions

White region:

$$X|W \sim N(4,4), \quad \sigma_1 = 2.$$

Black region:

$$X|B \sim N(6,9), \quad \sigma_2 = 3.$$

$$P(B) = \alpha, \quad P(W) = 1 - \alpha.$$

Condition:

$$P(B|X=5) = \frac{1}{2}.$$

Bayes' theorem:

$$P(B|X=5) = \frac{\alpha f(5|B)}{\alpha f(5|B) + (1-\alpha) f(5|W)}.$$

Normal density:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

$$f(5|B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}, \quad f(5|W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}.$$

Substitute:

$$\frac{3\alpha}{3\alpha + 2(1-\alpha)e^{-5/72}} = \frac{1}{2}.$$

Using $e^{-5/72} \approx 0.93$:

$$6\alpha = 3\alpha + 2(1-\alpha)(0.93).$$

$$3\alpha = 1.86 - 1.86\alpha.$$

$$4.86\alpha = 1.86.$$

$$\alpha \approx 0.3827.$$

## Q3. Minimizing Expected Distance

### (a) $X \sim$ **Uniform**$(0, A)$

$$f_X(x) = \frac{1}{A}.$$

$$E[|X-a|] = \int_0^a (a-x)\frac{1}{A}dx + \int_a^A (x-a)\frac{1}{A}dx.$$

Result:

$$\frac{2a^2 - 2aA + A^2}{2A}.$$

Differentiate:

$$\frac{1}{2A}(4a - 2A) = 0.$$

$$a = \frac{A}{2}.$$

### (b) $X \sim$ **Exponential**$(\lambda)$

$$f_X(x) = \lambda e^{-\lambda x}.$$

Derivative condition:

$$\frac{d}{da}E[|X-a|] = 1 - 2e^{-\lambda a} = 0.$$

$$e^{-\lambda a} = \frac{1}{2}.$$

$$a = \frac{\ln 2}{\lambda}.$$

## Q4. Mixture of Exponentials

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}.$$

$$P(X > t+s) = p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}.$$

$$P(X > t+s | X > t) = \frac{p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}.$$

## Q5. Conditional Distribution of Poisson Variables

$$X \sim \text{Poisson}(\lambda_1), \quad Y \sim \text{Poisson}(\lambda_2).$$

$$P(X = k | X + Y = n) = \binom{n}{k}\left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^{n-k}.$$

# Q6. Transformation of Random Variables

$$X \sim \text{Uniform}[-2,2], \quad f_X(x) = \frac{1}{4}.$$

$$g(x) = \begin{cases} x, & x \in [-2,-1], \\ 0, & x \in (-1,1), \\ x, & x \in [1,2]. \end{cases}$$

## CDF

$$F_Y(y) = 0, \quad y < -2.$$

$$F_Y(y) = \frac{y+2}{4}, \quad -2 \le y < -1.$$

$$F_Y(y) = \frac{1}{4}, \quad -1 \le y < 0.$$

$$P(Y=0) = \frac{1}{2}.$$

$$F_Y(0) = \frac{3}{4}.$$

$$F_Y(y) = \frac{3}{4}, \quad 0 < y < 1.$$

$$F_Y(y) = \frac{y+2}{4}, \quad 1 \le y \le 2.$$

$$F_Y(y) = 1, \quad y > 2.$$

## PDF

$$f_Y(y) = \frac{1}{4}, \quad -2 < y < -1.$$

$$f_Y(y) = \frac{1}{4}, \quad 1 < y < 2.$$

$$P(Y=0) = \frac{1}{2}.$$

## Q7. Binomial Stock Model

$$S_{1000} = S_0 u^k d^{1000-k}.$$

Condition:

$$k(\ln u - \ln d) + 1000 \ln d \geq \ln(1.30).$$

$$k \geq 470.$$

$$k \sim \text{Binomial}(1000, 0.52).$$

$$P(k \geq 470) = \sum_{k=470}^{1000} \binom{1000}{k} (0.52)^k (0.48)^{1000-k}.$$

## Q8. Exponential Repair Time

$$X \sim \text{Exponential}\left(\frac{1}{2}\right).$$

$$P(X > 2) = e^{-1}.$$

$$P(X \geq 10 | X > 9) = e^{-1/2}.$$

## Q9. Distribution of $Z = \sqrt{X^2 + Y^2}$

$$F_Z(z) = \int_{-z}^{z} \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y)\, dy\, dx.$$

$$f_Z(z) = z \int_{-z}^{z} \frac{f_X(x) \left[ f_Y(\sqrt{z^2-x^2}) + f_Y(-\sqrt{z^2-x^2}) \right]}{\sqrt{z^2-x^2}} dx.$$

## Q10. Joint Distribution of $M_n$ and $M_{n+1}$

$$M_n = \max(X_1, \ldots, X_n).$$

$$P(M_n \leq x) = F(x)^n.$$

Joint CDF:

$$P(M_n \leq x, M_{n+1} \leq y) = \begin{cases} F(x)^n F(y), & x \leq y, \\ F(y)^{n+1}, & x > y. \end{cases}$$